

Plan Switch Analysis

CS24BTECH11037 CS24BTECH11036

Query 1 Analysis

Table 1: Query 1 Plan Switch Execution Times and Classification

Sw.	Dates ($q_i \rightarrow q_j$)	$RT(P_i, q_i)$	$RT(P_j, q_j)$	$RT(P_i, q_j)$	$RT(P_j, q_i)$	Class	Justification
1	1992-01-07 to 08	298.69	1208.06	592.78	1007.43	EARLY	$RT(P_i, q_j) < RT(P_j, q_j)$; P_i is still better at q_j
2	1992-01-27 to 28	9260.92	3615.40	9812.73	2434.89	DELAYED	$RT(P_j, q_i) < RT(P_i, q_i)$; P_j was already better at q_i
3	1992-03-29 to 30	16052.16	10349.25	13593.91	8710.86	DELAYED	$RT(P_j, q_i) < RT(P_i, q_i)$; P_j was already better at q_i

Query 1 Hints Used

Switch 1

```
Pi: /** BitmapScan(lineitem) */  
Pj: /** IndexScan(lineitem idx_lineitem_shipdate) */
```

Switch 2

```
Pi: /** IndexScan(lineitem idx_lineitem_shipdate) */  
Pj: /** BitmapScan(lineitem) Parallel(lineitem 4) */
```

Switch 3

```
Pi: /** BitmapScan(lineitem) Parallel(lineitem 4) */  
Pj: /** SeqScan(lineitem) Parallel(lineitem 4) */
```

Query 2 Analysis

Table 2: Query 2 Plan Switch Execution Times and Classification

Sw.	Dates ($q_i \rightarrow q_j$)	RT(P_i, q_i)	RT(P_j, q_j)	RT(P_i, q_j)	RT(P_j, q_i)	Class	Justification
1	1992-01-01 to 02	1.83	683.38	2514.81	1.84	CORRECT	RT(P_i, q_i) < RT(P_j, q_i) and RT(P_j, q_j) < RT(P_i, q_j)
2	1992-02-18 to 19	17741.77	13225.83	11968.66	9814.43	DELAYED	RT(P_j, q_i) < RT(P_i, q_i); P_j was already better at q_i
3	1992-03-17 to 18	11429.44	17585.78	9797.26	14085.65	EARLY	RT(P_i, q_j) < RT(P_j, q_j); P_i is still better at q_j
4	1992-05-17 to 18	14095.43	10512.62	11457.47	9943.81	DELAYED	RT(P_j, q_i) < RT(P_i, q_i); P_j was already better at q_i
5	1992-05-18 to 19	10439.45	13216.30	9872.22	10895.88	EARLY	RT(P_i, q_j) < RT(P_j, q_j); P_i is still better at q_j
6	1997-12-24 to 25	14828.47	14096.33	12874.93	13005.17	DELAYED	RT(P_j, q_i) < RT(P_i, q_i); P_j was already better at q_i

Query 2 Hints Used

Switch 1

```
Pi: /*+ NestLoop(orders lineitem) IndexScan(orders idx_orders_orderdate)
      IndexScan(lineitem lineitem_pkey) */
Pj: /*+ NestLoop(orders lineitem) BitmapScan(orders) Parallel(orders 4)
      IndexScan(lineitem lineitem_pkey) */
```

Switch 2

```
Pi: /** NestLoop(orders lineitem) BitmapScan(orders) Parallel(orders 4)
      IndexScan(lineitem lineitem_pkey) */
Pj: /** NestLoop(orders lineitem) SeqScan(orders) Parallel(orders 4)
      IndexScan(lineitem lineitem_pkey) */
```

Switch 3

```
Pi: /** NestLoop(orders lineitem) SeqScan(orders) Parallel(orders 4)
      IndexScan(lineitem lineitem_pkey) */
Pj: /** NestLoop(orders lineitem) BitmapScan(orders) Parallel(orders 4)
      IndexScan(lineitem lineitem_pkey) */
```

Switch 4

```
Pi: /** NestLoop(orders lineitem) BitmapScan(orders) Parallel(orders 4)
      IndexScan(lineitem lineitem_pkey) */
Pj: /** NestLoop(orders lineitem) SeqScan(orders) Parallel(orders 4)
      IndexScan(lineitem lineitem_pkey) */
```

Switch 5

```
Pi: /** NestLoop(orders lineitem) SeqScan(orders) Parallel(orders 4)
      IndexScan(lineitem lineitem_pkey) */
Pj: /** HashJoin(orders lineitem) SeqScan(orders) Parallel(orders 4)
      SeqScan(lineitem) Parallel(lineitem 4) */
```

Switch 6

```
Pi: /** HashJoin(orders lineitem) SeqScan(orders) Parallel(orders 4)
      SeqScan(lineitem) Parallel(lineitem 4) */
Pj: /** HashJoin(orders lineitem) SeqScan(orders) Parallel(orders 4)
      SeqScan(lineitem) Parallel(lineitem 4) */
```

Query 3 Analysis

Query 3 Hints Used

Switch 1

```
Pi: /** NestLoop(o c 1) NestLoop(o c) BitmapScan(o) Parallel(o 4)
      IndexScan(c customer_pkey) IndexScan(1 lineitem_pkey) */
Pj: /** NestLoop(c o 1) HashJoin(c o) SeqScan(c) Parallel(c 4)
      BitmapScan(o) Parallel(o 4) IndexScan(1 lineitem_pkey) */
```

Switch 2

```
Pi: /** NestLoop(c o 1) HashJoin(c o) SeqScan(c) Parallel(c 4)
      BitmapScan(o) Parallel(o 4) IndexScan(1 lineitem_pkey) */
Pj: /** NestLoop(o c 1) HashJoin(o c) BitmapScan(o) Parallel(o 4)
      SeqScan(c) Parallel(c 4) IndexScan(1 lineitem_pkey) */
```

Switch 3

```
Pi: /** NestLoop(o c 1) HashJoin(o c) BitmapScan(o) Parallel(o 4)
      SeqScan(c) Parallel(c 4) IndexScan(1 lineitem_pkey) */
Pj: /** HashJoin(1 o c) HashJoin(1 o) SeqScan(1) Parallel(1 4)
      BitmapScan(o) Parallel(o 4) SeqScan(c) Parallel(c 4) */
```

Switch 4

```
Pi: /** HashJoin(1 o c) HashJoin(1 o) SeqScan(1) Parallel(1 4)
      BitmapScan(o) Parallel(o 4) SeqScan(c) Parallel(c 4) */
Pj: /** HashJoin(1 o c) HashJoin(1 o) SeqScan(1) Parallel(1 4)
      SeqScan(o) Parallel(o 4) SeqScan(c) Parallel(c 4) */
```

Switch 5

```
Pi: /** HashJoin(1 o c) HashJoin(1 o) SeqScan(1) Parallel(1 4)
      SeqScan(o) Parallel(o 4) SeqScan(c) Parallel(c 4) */
Pj: /** HashJoin(1 o c) SeqScan(1) Parallel(1 4) HashJoin(o c)
      SeqScan(o) Parallel(o 4) SeqScan(c) Parallel(c 4) */
```

Switch 6

```
Pi: /** HashJoin(1 o c) SeqScan(1) Parallel(1 4) HashJoin(o c)
      SeqScan(o) Parallel(o 4) SeqScan(c) Parallel(c 4) */
Pj: /** HashJoin(1 o c) SeqScan(1) Parallel(1 4) HashJoin(o c)
      BitmapScan(o) Parallel(o 4) SeqScan(c) Parallel(c 4) */
```

Switch 7

```
Pi: /** HashJoin(1 o c) SeqScan(1) Parallel(1 4) HashJoin(o c)
      BitmapScan(o) Parallel(o 4) SeqScan(c) Parallel(c 4) */
Pj: /** HashJoin(1 o c) SeqScan(1) Parallel(1 4) HashJoin(o c)
      SeqScan(o) Parallel(o 4) SeqScan(c) Parallel(c 4) */
```

Switch 8

```
Pi: /** HashJoin(l o c) SeqScan(l) Parallel(1 4) HashJoin(o c)
      SeqScan(o) Parallel(o 4) SeqScan(c) Parallel(c 4) */
Pj: /** HashJoin(l o c) HashJoin(l o) SeqScan(l) Parallel(1 4)
      SeqScan(o) Parallel(o 4) SeqScan(c) Parallel(c 4) */
```

Switch 9

```
Pi: /** HashJoin(l o c) HashJoin(l o) SeqScan(l) Parallel(1 4)
      SeqScan(o) Parallel(o 4) SeqScan(c) Parallel(c 4) */
Pj: /** HashJoin(l o c) SeqScan(l) Parallel(1 4) HashJoin(o c)
      SeqScan(o) Parallel(o 4) SeqScan(c) Parallel(c 4) */
```

Table 3: Query 3 Plan Switch Execution Times and Classification

Sw.	Dates ($q_i \rightarrow q_j$)	$RT(P_i, q_i)$	$RT(P_j, q_j)$	$RT(P_i, q_j)$	$RT(P_j, q_i)$	Class	Justification
1	1992-01-03 to 04	1109.67	1551.44	1149.56	977.67	DELAYED	$RT(P_j, q_i) < RT(P_i, q_i)$; P_j was already better at q_i
2	1992-01-06 to 07	2483.49	2181.82	1928.74	1679.61	DELAYED	$RT(P_j, q_i) < RT(P_i, q_i)$; P_j was already better at q_i
3	1992-01-17 to 18	3555.31	13406.80	4370.75	10988.60	EARLY	$RT(P_i, q_j) < RT(P_j, q_j)$; P_i is still better at q_j
4	1992-02-16 to 17	14371.07	12046.44	12460.70	10158.96	DELAYED	$RT(P_j, q_i) < RT(P_i, q_i)$; P_j was already better at q_i
5	1992-03-13 to 14	11891.37	11994.96	10237.31	10184.39	DELAYED	$RT(P_j, q_i) < RT(P_i, q_i)$; P_j was already better at q_i
6	1992-03-19 to 20	12040.55	13891.80	10215.65	12181.70	EARLY	$RT(P_i, q_j) < RT(P_j, q_j)$; P_i is still better at q_j
7	1992-05-20 to 21	13205.25	11914.48	11412.97	10607.52	DELAYED	$RT(P_j, q_i) < RT(P_i, q_i)$; P_j was already better at q_i
8	1992-12-04 to 05	12005.06	14192.18	10404.60	10513.35	DELAYED	$RT(P_j, q_i) < RT(P_i, q_i)$; P_j was already better at q_i
9	1993-03-07 to 08	14234.08	14366.60	10419.70	10416.57	DELAYED	$RT(P_j, q_i) < RT(P_i, q_i)$; P_j was already better at q_i